Type MS · Explain how the pupil responds to changes in light intensity

Answer 1 of 4 · Question 6 part b.i · 5 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

6 (a)	C (fovea); A/B /D have no light sensitive cells		1
(b) (i)	Any 5 from • light becomes less/becomes darker; • reference to dim light detected by retina; • radial muscles in iris contract; • circular muscles relax; • widen/increase diameter of pupil; • to keep a constant amount of light entering the eye	Allow reference to B being in dimmer light allow more light entering the eye	5
(ii)	• occurs when bright light is shone into eye; • too much light could damage retina; • occurs quickly to protect eye/prevent too much light entering;		3

Total 9 marks

Type MS · Explain how the pupil responds to changes in light intensity

Answer 1 of 4 · Question 6 part b.i · 5 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the pupil responds to changes in light intensity

Answer 2 of 4 · Question 6 part b.ii · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number				
6	(a)	C (fovea); A/B /D have no light sensitive cells		1
	(b) (i)	Any 5 from • light becomes less/becomes darker; • reference to dim light detected by retina; • radial muscles in iris contract; • circular muscles relax; • widen/increase diameter of pupil; • to keep a constant amount of light entering the eye	Allow reference to B being in dimmer light allow more light entering the eye	5
	(ii)	• occurs when bright light is shone into eye; • too much light could damage retina; • occurs quickly to protect eye/prevent too much light entering;		3

Total 9 marks

Type MS · Explain how the pupil responds to changes in light intensity

Answer 2 of 4 · Question 6 part b.ii · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how the pupil responds to changes in light intensity

Answer 3 of 4 · Question 6 part c.i · 3 marks

2024 JUN · Paper 02 · find part c.i in this mark scheme extract

6 (a)	• dim room or bright room/lamp at different distances; • allow time for (pupil) to adjust; • use a camera/video to take a picture/recording of the eye/ use of ruler; • measure the width/diameter of the pupil;	Ignore change light intensity	(3)
(b)	• diameter of pupil 6- 8 mm; • substitution 3.14 x radius of pupil; • accept any value between 28-51 mm^2;	If the diameter is used instead of the radius, then 2 marks can be awarded for any value between 113-201mm^2 Any value between 28-51 scores 3 marks	(3)
(c) (i)	• circular muscles relax; • radial muscles contract; • pupil gets larger/wider /dilates/ increase in size;		(3)
(ii)	• to allow more light to enter the eye; • to see image more clearly/better;		(2)

Total for Question 6 = 11 marks

Type MS · Explain how the pupil responds to changes in light intensity

Answer 4 of 4 · Question 6 part c.ii · 2 marks

2024 JUN · Paper 02 · find part c.ii in this mark scheme extract

6 (a)	• dim room or bright room/lamp at different distances; • allow time for (pupil) to adjust; • use a camera/video to take a picture/recording of the eye/ use of ruler; • measure the width/diameter of the pupil;	Ignore change light intensity	(3)
(b)	• diameter of pupil 6- 8 mm; • substitution 3.14 x radius of pupil; • accept any value between 28-51 mm^2;	If the diameter is used instead of the radius, then 2 marks can be awarded for any value between 113-201mm^2 Any value between 28-51 scores 3 marks	(3)
(c) (i)	• circular muscles relax; • radial muscles contract; • pupil gets larger/wider /dilates/ increase in size;		(3)
(ii)	• to allow more light to enter the eye; • to see image more clearly/better;		(2)

Total for Question 6 = 11 marks